

A Shifted-Beam Method for Near-Field Wireless Power Transfer using Parasitic Arrays

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Abstract—Optimizing near field (NF) wireless power transfer (WPT) to ensure maximum power transfer efficiency (PTE) is essential for a number of applications such as radio-frequency identification (RFID) and device charging. A new WPT method using a parasitic array is proposed, inspired by a shifted-beam (SB) wave interference technique for field focusing. The SB method is implemented for target distances between 1λ and 2λ at 2.4 GHz ($\lambda=12.5$ cm), which lie roughly in the radiative NF.

PTE from the SB method is compared with results from a uniformly excited array of similar length, as well as a phase optimization (PO) method based on conjugate-phase (CP) focusing. The transmit (T_x) system is implemented as an array of centered dipoles, with a 0.48λ non-excited dipole used as the receive (R_x) element. The results show an enhancement of PTE compared to the baseline uniform array, and in some cases the PO array. The simplicity of the SB method makes it an advantageous design choice.

Index Terms—Arrays, dipoles, field focusing, wireless power transfer, optimisation, shifted beam, conjugate phase, parasitic.

I. INTRODUCTION

WIRELESS power transfer (WPT) is useful for powering devices at a distance where mobility is important, or if the device is in a difficult-to-reach location. The convenience of WPT has therefore made it a major area of research interest. Different mechanisms for achieving WPT are viable depending on the frequency, distance, power and efficiency requirements. Using antenna arrays for WPT in the near field (NF) and far field (FF) is also possible [1].

The goals of WPT are typically to maximize transmission distance, power transmission efficiency (PTE) and/or transmitted power; this paper focuses on maximizing PTE. In this paper a new WPT method using a parasitic array is proposed, inspired by a shifted-beam (SB) wave interference technique for field focusing. The PTE is compared with results from a technique based on conjugate-phase (CP) field focusing using phase optimization (PO), as well as a uniformly excited array.

II. WIRELESS POWER TRANSFER SYSTEM SETUP

Dipole antennas are simple and easily realized, with moderate gain [2]. A linear array of center-fed dipoles can be used for WPT; by adjusting the dipole lengths, inter-dipole spacing, excitation amplitude and/or phase, PTE can be optimized (fig. 1). The setups in this paper are modelled using the Hallén-Pocklington equation (HPE) for 1D wires in MATLAB R2021b [3]. The T_x elements are excited using a delta-gap excitation of 1V at the center. The R_x element is a single

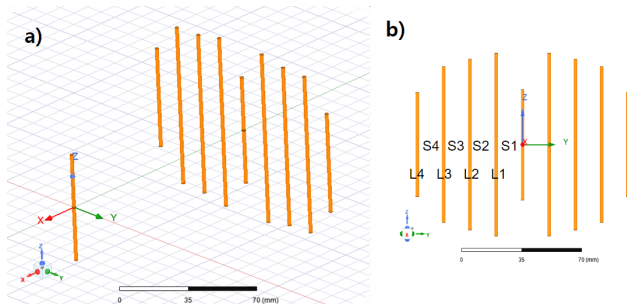


Fig. 1. (a) Example of WPT setup with T_x array and R_x element. (b) Labelling of element lengths ($L_1 \dots L_n$) and inter-element spacing ($S_1 \dots S_n$) for the T_x array, used as optimization parameters for SB-WPT.

non-excited 0.48λ dipole placed at the desired target distance. PTE is defined as the ratio of the received and transmitted power, i.e. $P = I^2 Z$. P_{T_x} is calculated using the current at the center of all excited T_x dipoles, and P_{R_x} is calculated using the induced current at the center of the R_x dipole.

The array designs are determined using global optimization in MATLAB ("fmincon" function). Comparison with results from finite-element method (FEM) solvers such as COMSOL 5.6 (RF module) shows good agreement provided that the dipole radii are equal. However, dipole radii of $\sim 0.01\lambda$ ($r \sim 1$ mm at 2.4 GHz) leads to numeric instability for the HPE when using a pulse basis [3]. To prevent this issue, $r=10^{-6}\lambda$ is used. Each dipole is divided into 41 discretizations regardless of length. For physical realization, the MATLAB results can be further optimized in a FEM solver such as COMSOL or Ansys HFSS to create a design that can be fabricated.

III. WPT WITH SHIFTED-BEAM INSPIRED ARRAYS

The SB method uses wave interference to improve NF field focusing. In the NF, antenna fields from array elements exhibit a spatial shift corresponding to their relative displacement (as opposed to the FF where the effect would be a phase shift) [4]. Defining the weights appropriately allows the width of the field spot at the focal plane to be substantially decreased. For a parasitically excited T_x array where only the center element is excited, field characteristics can be tuned by adjusting the inter-dipole spacing and dipole lengths (and thus their input impedance) for maximum PTE. Global optimization is used to account for mutual coupling, as the element spacing may be highly sub-wavelength.

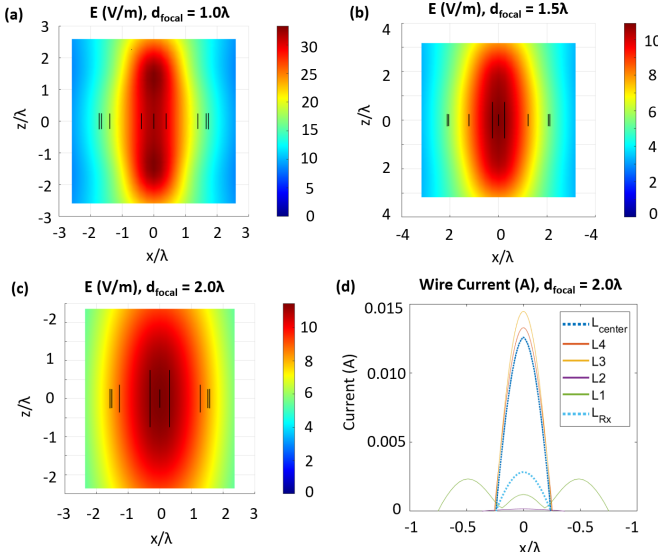


Fig. 2. (a, b, c) Optimized 9 element arrays for maximum PTE using SB method, with E field magnitude at plane of R_x element. (d) Currents along lengths of T_x array dipoles, corresponding to fig. 2c. Dipoles are labelled according to fig. 1b.

The optimization starts with all dipole elements set to a length of 0.48λ and spaced $\lambda/2$ apart; the range of allowable dipole lengths and spacings is $[\frac{1}{5}, \frac{3}{2}]\lambda$ and $[\frac{1}{20}, 1]\lambda$ respectively (fig. 1). The excited center dipole is fixed at a length of 0.48λ to maximize radiation. The optimal parameters for a 5 and 9 element array are listed in table I; array size in all cases is $\sim \frac{N\lambda}{2}$, where N is the number of parasitically excited dipoles on each side of the center element. Field amplitude is calculated using NF radiation equations (fig. 2) [5].

IV. WPT WITH PHASE OPTIMIZATION

To enhance PTE, one common approach is to adjust the phases of the elements such that the wavefronts add constructively at the focal point (i.e. the location of the R_x element); this is the premise of CP focusing [6]. It has the benefit of only needing the displacement between the T_x and R_x elements for implementation, although all elements must be excited using a feed network. The accumulated phase between the R_x element and any array element is:

$$\phi = \frac{2\pi}{\lambda} |r_{\text{focal}} - r_{\text{elem}}| \quad (1)$$

When the elements are spaced $\sim 1\lambda$ apart in a large array, mutual coupling is low and can be ignored [6]. In the case of $\lambda/2$ element spacing used in this paper, PO can be used to adjust the phases and maximize PTE. All T_x array elements are excited with 1V at the center. The optimized phases and PTE values are listed in table I and II respectively.

V. CONCLUSIONS

A WPT method inspired by SB focusing using parasitic dipole arrays is studied for distances between 1λ and 2λ . The results show considerable PTE improvement in many cases

TABLE I
OPTIMIZED PARAMETERS (LABELLED IN FIG. 1) FOR SB AND PO WPT METHOD, FOR 5 AND 9 ELEMENT ARRAYS. MATLAB "FMINCON" OPTIMALITY TOLERANCE IS 10^{-6} .

$d_{R_x} (\lambda)$	Dipole Lengths 5, 9 elems. $L_1 \dots L_n (\lambda)$	Dipole Spacings 5, 9 elems. $S_1 \dots S_n (\lambda)$	Dipole Phases 5, 9 elems. (rad)
1.0	(0.50, 0.49), (0.49, 0.48, 0.49, 0.50)	(0.34, 0.70), (0.39, 1.00, 0.26, 0.072)	(1.027, 3.14), (0.73, 3.14, -0.42, 2.10)
1.5	(0.49, 0.49), (1.50, 0.50, 0.49, 0.50)	(0.67, 0.59), (0.24, 0.97, 0.83, 0.06)	(0.83, 2.28), (0.37, 2.09, -1.90, 0.60)
2.0	(0.94, 0.49), (1.5, 0.72, 0.49, 0.50)	(0.46, 0.84), (0.30, 0.96, 0.24, 0.056)	(0.66, 1.80), (0.28, 1.51, -2.68, -0.68)

TABLE II
COMPARISON OF PTE (%) FOR 5 AND 9 ELEMENT ARRAYS.

$d_{R_x} (\lambda)$	SB Method: 5, 9 elem. array ($\sim 2\lambda$, $\sim 4\lambda$ length)	PO Method: 5, 9 elem. array (2λ , 4λ length)	Uniform Array: 5, 9 elem. array (2λ , 4λ length)
1.0	15.4, 21.1	10.5, 12.3	3.0, 1.8
1.5	6.8, 8.5	6.2, 8.8	3.0, 1.3
2.0	2.7, 5.0	4.0, 5.8	2.4, 0.7

over a uniformly excited array of 0.48λ dipoles spaced $\lambda/2$ apart. Compared to PO, the SB inspired method achieves greater PTE up to $\sim 2\lambda$ in this paper (table II). At shorter distances, approaching the boundary of the radiative NF, PO is less effective than the SB method. One advantage of SB inspired WPT is its simplicity, as only the center element is excited and no feed network is required.

The SB inspired method can be further optimized for charging multiple devices at a distance, as well as RFID applications. The concept bears similarity to other work on beam shaping using impedance-loaded metawire gratings, which represents a direction for future work [7]. Loading the dipoles as well as adding optimization constraints on sensitivity and array length can further improve the design.

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